

| <table><tr><th colspan="2">Table 2</th></tr><tr><td>Irene Blanco<br/>Casallas</td><td>William Nimtz</td></tr><tr><td></td><td>Sujal Vajire</td></tr></table> |                                                            | Table 2 |  | Irene Blanco<br>Casallas | William Nimtz   |          | Sujal Vajire | <table><tr><th colspan="2">Table 3</th></tr><tr><td>Max Hortop</td><td>Lautaro Garcia</td></tr><tr><td>Dashiel Matlock</td><td>Lin Tian</td></tr></table> |  | Table 3                                                                                                                                                                |  | Max Hortop | Lautaro Garcia | Dashiel Matlock | Lin Tian    | <table><tr><th colspan="2">Table 4</th></tr><tr><td>Jessica Duran</td><td>Ethan Kepros<br/>Mohamed<br/>Abdulazim Ali Afifi El<br/>Habet</td></tr><tr><td>Mi Zhou</td><td></td></tr></table> |                     | Table 4                                                                                                                                                               |  | Jessica Duran | Ethan Kepros<br>Mohamed<br>Abdulazim Ali Afifi El<br>Habet | Mi Zhou        |                   | <table><tr><th colspan="2">Table 5</th></tr><tr><td>Sam Hu</td><td>Jack Bajcz</td></tr><tr><td>Basma AlMahmood</td><td>Samuel Goodluck</td></tr></table> |                 | Table 5 |  | Sam Hu | Jack Bajcz | Basma AlMahmood | Samuel Goodluck |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------|---------|--|--------------------------|-----------------|----------|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|--|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|------------|----------------|-----------------|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|---------------|------------------------------------------------------------|----------------|-------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|---------|--|--------|------------|-----------------|-----------------|
| Table 2                                                                                                                                                      |                                                            |         |  |                          |                 |          |              |                                                                                                                                                           |  |                                                                                                                                                                        |  |            |                |                 |             |                                                                                                                                                                                             |                     |                                                                                                                                                                       |  |               |                                                            |                |                   |                                                                                                                                                          |                 |         |  |        |            |                 |                 |
| Irene Blanco<br>Casallas                                                                                                                                     | William Nimtz                                              |         |  |                          |                 |          |              |                                                                                                                                                           |  |                                                                                                                                                                        |  |            |                |                 |             |                                                                                                                                                                                             |                     |                                                                                                                                                                       |  |               |                                                            |                |                   |                                                                                                                                                          |                 |         |  |        |            |                 |                 |
|                                                                                                                                                              | Sujal Vajire                                               |         |  |                          |                 |          |              |                                                                                                                                                           |  |                                                                                                                                                                        |  |            |                |                 |             |                                                                                                                                                                                             |                     |                                                                                                                                                                       |  |               |                                                            |                |                   |                                                                                                                                                          |                 |         |  |        |            |                 |                 |
| Table 3                                                                                                                                                      |                                                            |         |  |                          |                 |          |              |                                                                                                                                                           |  |                                                                                                                                                                        |  |            |                |                 |             |                                                                                                                                                                                             |                     |                                                                                                                                                                       |  |               |                                                            |                |                   |                                                                                                                                                          |                 |         |  |        |            |                 |                 |
| Max Hortop                                                                                                                                                   | Lautaro Garcia                                             |         |  |                          |                 |          |              |                                                                                                                                                           |  |                                                                                                                                                                        |  |            |                |                 |             |                                                                                                                                                                                             |                     |                                                                                                                                                                       |  |               |                                                            |                |                   |                                                                                                                                                          |                 |         |  |        |            |                 |                 |
| Dashiel Matlock                                                                                                                                              | Lin Tian                                                   |         |  |                          |                 |          |              |                                                                                                                                                           |  |                                                                                                                                                                        |  |            |                |                 |             |                                                                                                                                                                                             |                     |                                                                                                                                                                       |  |               |                                                            |                |                   |                                                                                                                                                          |                 |         |  |        |            |                 |                 |
| Table 4                                                                                                                                                      |                                                            |         |  |                          |                 |          |              |                                                                                                                                                           |  |                                                                                                                                                                        |  |            |                |                 |             |                                                                                                                                                                                             |                     |                                                                                                                                                                       |  |               |                                                            |                |                   |                                                                                                                                                          |                 |         |  |        |            |                 |                 |
| Jessica Duran                                                                                                                                                | Ethan Kepros<br>Mohamed<br>Abdulazim Ali Afifi El<br>Habet |         |  |                          |                 |          |              |                                                                                                                                                           |  |                                                                                                                                                                        |  |            |                |                 |             |                                                                                                                                                                                             |                     |                                                                                                                                                                       |  |               |                                                            |                |                   |                                                                                                                                                          |                 |         |  |        |            |                 |                 |
| Mi Zhou                                                                                                                                                      |                                                            |         |  |                          |                 |          |              |                                                                                                                                                           |  |                                                                                                                                                                        |  |            |                |                 |             |                                                                                                                                                                                             |                     |                                                                                                                                                                       |  |               |                                                            |                |                   |                                                                                                                                                          |                 |         |  |        |            |                 |                 |
| Table 5                                                                                                                                                      |                                                            |         |  |                          |                 |          |              |                                                                                                                                                           |  |                                                                                                                                                                        |  |            |                |                 |             |                                                                                                                                                                                             |                     |                                                                                                                                                                       |  |               |                                                            |                |                   |                                                                                                                                                          |                 |         |  |        |            |                 |                 |
| Sam Hu                                                                                                                                                       | Jack Bajcz                                                 |         |  |                          |                 |          |              |                                                                                                                                                           |  |                                                                                                                                                                        |  |            |                |                 |             |                                                                                                                                                                                             |                     |                                                                                                                                                                       |  |               |                                                            |                |                   |                                                                                                                                                          |                 |         |  |        |            |                 |                 |
| Basma AlMahmood                                                                                                                                              | Samuel Goodluck                                            |         |  |                          |                 |          |              |                                                                                                                                                           |  |                                                                                                                                                                        |  |            |                |                 |             |                                                                                                                                                                                             |                     |                                                                                                                                                                       |  |               |                                                            |                |                   |                                                                                                                                                          |                 |         |  |        |            |                 |                 |
| <table><tr><th colspan="2">Table 1</th></tr><tr><td>Abraham Rai</td><td>Alexander Lover</td></tr><tr><td>Xin Dong</td><td>Jingyi Ge</td></tr></table>        |                                                            | Table 1 |  | Abraham Rai              | Alexander Lover | Xin Dong | Jingyi Ge    |                                                                                                                                                           |  | <table><tr><th colspan="2">Table 7</th></tr><tr><td>Saksham Mamtani</td><td>Andrew Tran</td></tr><tr><td>Sanjida Meherin</td><td>Aidana Bekbulatkyzy</td></tr></table> |  | Table 7    |                | Saksham Mamtani | Andrew Tran | Sanjida Meherin                                                                                                                                                                             | Aidana Bekbulatkyzy | <table><tr><th colspan="2">Table 6</th></tr><tr><td>Madison Mercer</td><td>Md Shariful Islam</td></tr><tr><td>Grace Akintan</td><td>Michael Dittman</td></tr></table> |  | Table 6       |                                                            | Madison Mercer | Md Shariful Islam | Grace Akintan                                                                                                                                            | Michael Dittman |         |  |        |            |                 |                 |
| Table 1                                                                                                                                                      |                                                            |         |  |                          |                 |          |              |                                                                                                                                                           |  |                                                                                                                                                                        |  |            |                |                 |             |                                                                                                                                                                                             |                     |                                                                                                                                                                       |  |               |                                                            |                |                   |                                                                                                                                                          |                 |         |  |        |            |                 |                 |
| Abraham Rai                                                                                                                                                  | Alexander Lover                                            |         |  |                          |                 |          |              |                                                                                                                                                           |  |                                                                                                                                                                        |  |            |                |                 |             |                                                                                                                                                                                             |                     |                                                                                                                                                                       |  |               |                                                            |                |                   |                                                                                                                                                          |                 |         |  |        |            |                 |                 |
| Xin Dong                                                                                                                                                     | Jingyi Ge                                                  |         |  |                          |                 |          |              |                                                                                                                                                           |  |                                                                                                                                                                        |  |            |                |                 |             |                                                                                                                                                                                             |                     |                                                                                                                                                                       |  |               |                                                            |                |                   |                                                                                                                                                          |                 |         |  |        |            |                 |                 |
| Table 7                                                                                                                                                      |                                                            |         |  |                          |                 |          |              |                                                                                                                                                           |  |                                                                                                                                                                        |  |            |                |                 |             |                                                                                                                                                                                             |                     |                                                                                                                                                                       |  |               |                                                            |                |                   |                                                                                                                                                          |                 |         |  |        |            |                 |                 |
| Saksham Mamtani                                                                                                                                              | Andrew Tran                                                |         |  |                          |                 |          |              |                                                                                                                                                           |  |                                                                                                                                                                        |  |            |                |                 |             |                                                                                                                                                                                             |                     |                                                                                                                                                                       |  |               |                                                            |                |                   |                                                                                                                                                          |                 |         |  |        |            |                 |                 |
| Sanjida Meherin                                                                                                                                              | Aidana Bekbulatkyzy                                        |         |  |                          |                 |          |              |                                                                                                                                                           |  |                                                                                                                                                                        |  |            |                |                 |             |                                                                                                                                                                                             |                     |                                                                                                                                                                       |  |               |                                                            |                |                   |                                                                                                                                                          |                 |         |  |        |            |                 |                 |
| Table 6                                                                                                                                                      |                                                            |         |  |                          |                 |          |              |                                                                                                                                                           |  |                                                                                                                                                                        |  |            |                |                 |             |                                                                                                                                                                                             |                     |                                                                                                                                                                       |  |               |                                                            |                |                   |                                                                                                                                                          |                 |         |  |        |            |                 |                 |
| Madison Mercer                                                                                                                                               | Md Shariful Islam                                          |         |  |                          |                 |          |              |                                                                                                                                                           |  |                                                                                                                                                                        |  |            |                |                 |             |                                                                                                                                                                                             |                     |                                                                                                                                                                       |  |               |                                                            |                |                   |                                                                                                                                                          |                 |         |  |        |            |                 |                 |
| Grace Akintan                                                                                                                                                | Michael Dittman                                            |         |  |                          |                 |          |              |                                                                                                                                                           |  |                                                                                                                                                                        |  |            |                |                 |             |                                                                                                                                                                                             |                     |                                                                                                                                                                       |  |               |                                                            |                |                   |                                                                                                                                                          |                 |         |  |        |            |                 |                 |

In [ ]:

# CMSE 801 - Day 10

Planning and visualizing compartmental models

Feb 12, 2026



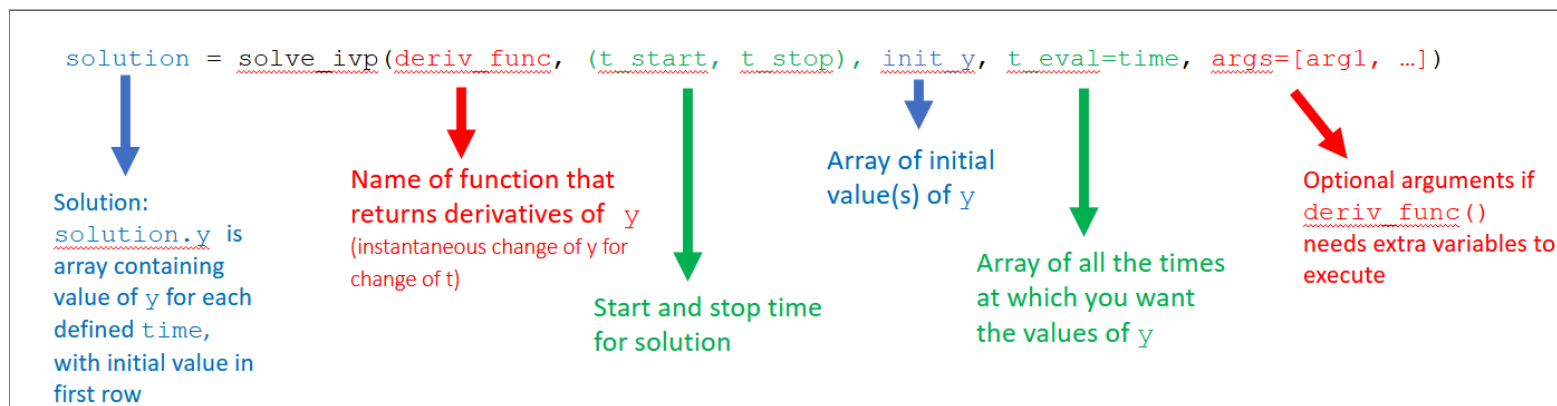
## General Course Announcements

- **Reminder:**
  - Homework 1 is due Saturday Feb 21.
  - Homework 2 is due Saturday Feb 28.
- Quiz 1 is **today** during the first 30 minutes of class.

## Questions/comments from Day 10 PCA

- Still running into some issues with using `solve_ivp` -- **we'll revisit some basic components of using `solve_ivp` in a moment.**
- Some questions about the specific "Viral Kinetics" example used in the PCA, so **we'll take a look at that**
  - Related: questions about making the 3D plot

Let's revisit how we use `solve_ivp` and specifically the `args` argument



If we want to pass in model parameters without having to change the `derivs` function, we can do that, but we need to define them in the function arguments

In [1]:

```
def derivs(time, curr_vals, A, m):  
  
    # Declare any unchanging parameters  
    g = -9.81 # m/s^2  
  
    # Unpack the current values of the variables we wish to "update" from the curr_vals list  
    h, v = curr_vals  
  
    # Right-hand side of odes, which are used to compute the derivative  
    dhdt = v  
    dvdt = g + (-0.65 * A * v * abs(v))/m  
  
    return dhdt, dvdt
```

Then, when we call `solve_ivp`, we need to pass these through via the `args` argument

In [2]:

```
import numpy as np
from scipy.integrate import solve_ivp

# Declare Variables for initial conditions
h0 = 2000 # meters ### MODIFIED CODE
v0 = 0 # m/s
g = -9.81 # m/s^2
tmax = 30 # seconds
dt = 0.1 # seconds

# Define the time array
time = np.arange(0, tmax + dt, dt)

# Store the initial values in a list
init = [h0, v0]

# Define the model parameters needed for the derivs function
A = 0.4 # m^2
m = 80.0 # kg

# Solve the odes with solve_ivp
sol = solve_ivp(derivs, (0, tmax), init, t_eval = time, args=(A, m))
```

And, of course, we need to them unpack the result to visualize it

In [4]:

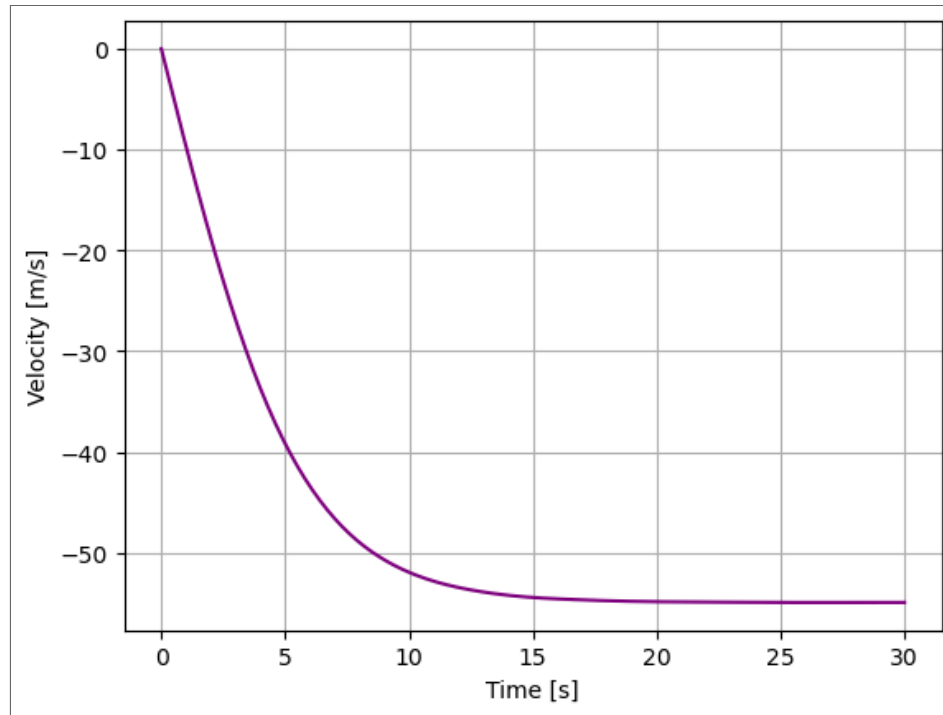
```
import matplotlib.pyplot as plt

#print(sol.y)

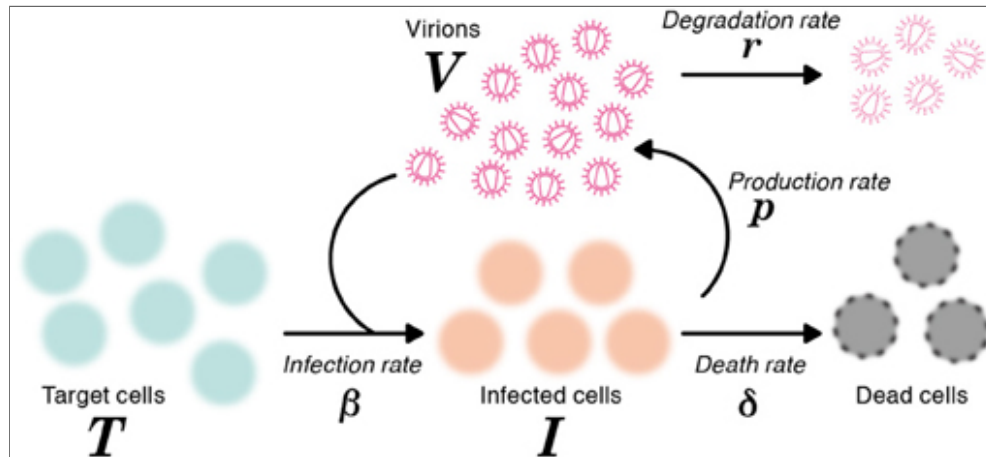
# Unpack the results stored in the solution variable, "sol"
h = sol.y[0,:]
v = sol.y[1,:]
t = sol.t

# Plot the results we unpacked from "sol" for velocity
plt.figure(2)
plt.plot(t, v, color = 'purple')
plt.xlabel('Time [s]')
plt.ylabel('Velocity [m/s]')
plt.grid()
```





Let's walk through the pieces for the viral kinetics problem



The model was:

$$\frac{dT}{dt} = -\beta TV,$$

$$\frac{dI}{dt} = \beta TV - \delta I,$$

$$\frac{dV}{dt} = pI - rV.$$

Remember: the order of our values and derivatives matters!!

In [5]:

```
# First thing I want to do is turn the equations into code
def derivs(t, curr_vals, beta, delta, p, r):
    # unpack list of incoming variables (which are stored in curr_vals)
    target = curr_vals[0]
    infected = curr_vals[1]
    virion = curr_vals[2]

    # use variables to compute derivatives on the right-hand-side of the above equations
    # example: dTdt = < equation from above >
    dTdt = -beta*target*virion
    dIdt = (beta*target*virion) - (delta*infected)

    dVdt = (p*infected) - (r*virion)

    # return the derivatives, one for each equation
    return dTdt, dIdt, dVdt
```

In [6]:

```
# Then I need to set up my initial conditions  
target_0 = 4e8  
infected_0 = 0.0  
virions_0 = 1.0  
# Here we put them all into a list so that they can be passed to solve_ivp  
initial = [target_0, infected_0, virions_0]
```

In [7]:

```
# Also need to define the times that I want my solution for  
time = np.linspace(0, 10., 1000)
```

In [8]:

```
# And define my model parameters  
beta = 1e-5  
delta = 4.0  
p = 1e-2  
r = 1
```

In [9]:

```
# finally, I can call solve_ivp and unpack my solutions
sol = solve_ivp(derivs, (time[0], time[-1]), initial,
                t_eval=time, args=(beta, delta, p, r))
tar = sol.y[0, :]
inf = sol.y[1, :]
vir = sol.y[2, :]
```

In [10]:

```
# With my solutions, I can visualize the result
#This initializes a figure object
plt.figure(figsize = (15,8))

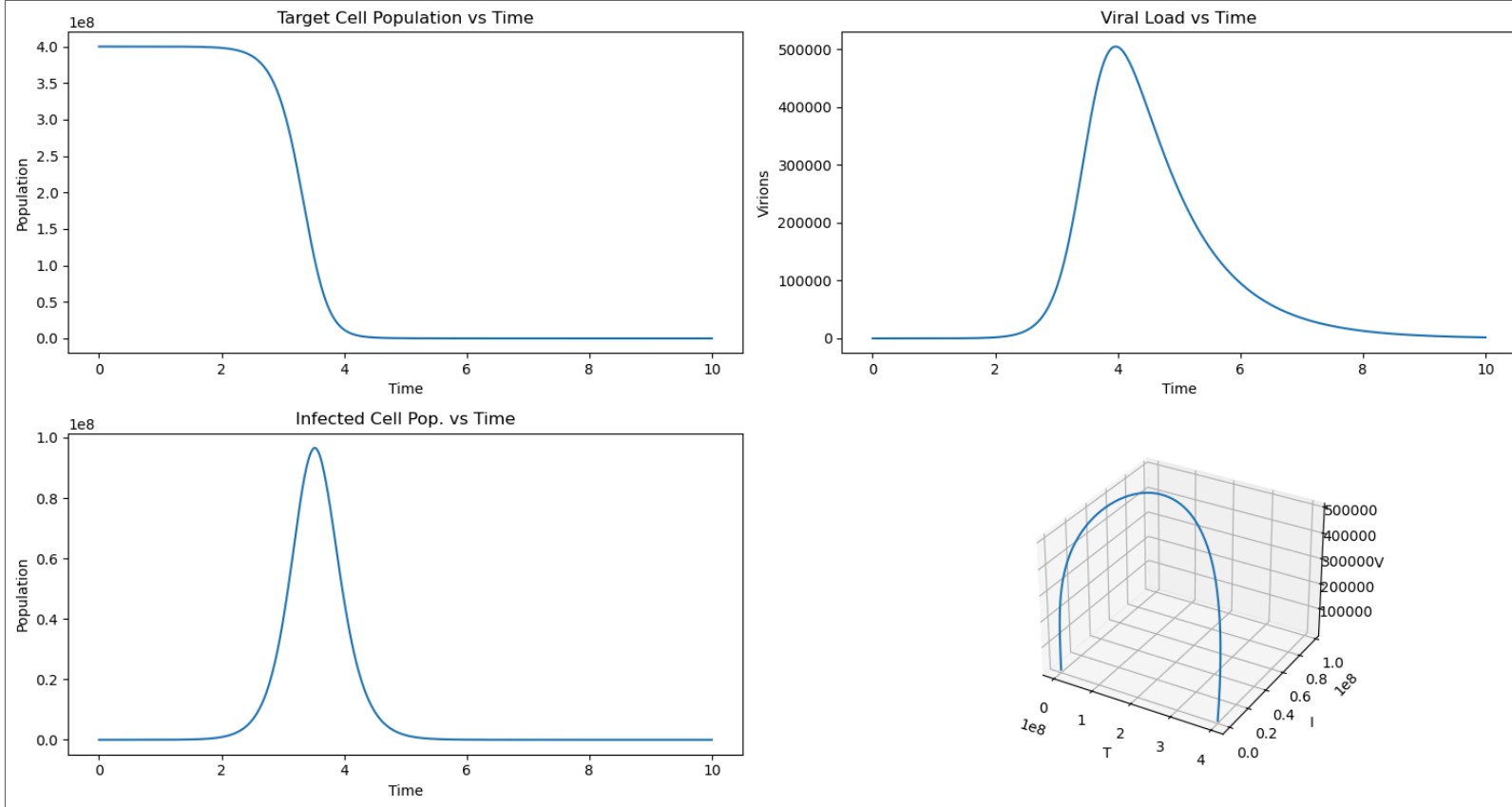
plt.subplot(2,2,1)
plt.plot(time,tar)
plt.title('Target Cell Population vs Time')
plt.xlabel('Time')
plt.ylabel('Population')

plt.subplot(2,2,2)
plt.plot(time, vir)
plt.title('Viral Load vs Time')
plt.xlabel('Time')
plt.ylabel('Virions')

plt.subplot(2,2,3)
plt.plot(time, inf)
plt.title('Infected Cell Pop. vs Time')
plt.xlabel('Time')
plt.ylabel('Population')

ax4 = plt.subplot(2,2,4, projection = '3d')
plt.plot(tar, inf, vir)
ax4.set_xlabel("T")
ax4.set_ylabel("I")
ax4.set_zlabel("V")

# Make sure to call this last. This command resizes the spaces between your plots to try to avoid overlap
plt.tight_layout()
```



## Goals for today

- Spend some time **not coding** and trying to carefully think through how you might build some compartmental models
- Think about what the compartments are and how they interact. **What defines how they evolve over time?** (i.e. what might their system of ODEs look like?)



Looking forward - Day 11 PCA: Look at another compartmental model example and practice using `solve_ivp` some more – modeling a zombie apocalypse.

- You'll explore a model for how a zombie outbreak might occur and evolve for a variety of model variations and parameters.
- You'll use `solve_ivp` some more.